Homework 7: DCGAN

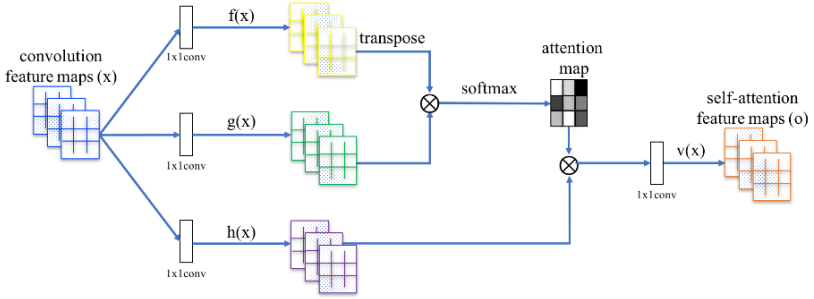
CS 2470

Due Dec 6th, 2019 at 11:59pm

1 CS2470-only Questions

1. The model you implemented for this assignment is built upon convolutional layers. A newly proposed model, authored by none other than Ian Goodfellow, is the Self-Attention Generative Adversarial Network (SAGAN), which has achieved state-of-the-art results in image generation tasks for multi-class datasets. You can find the paper here. What is the motivation for using self-attention in the image setting? How do the authors incorporate self-attention into their model?

   Answer: Regular GAN struggles with capturing geometric or structural patterns that occur consistently in some classes due to the heavily dependence on convolution. Long range dependencies can only be processed after passing through several convolutional layers. However, incorporating the self-attention module could help to model long range, multi-level dependencies across image regions. In terms of implementation, they added a self-attention module to get feature maps where f(x), g(x) and h(x) are used in the same fashion in self-attention RNN. They fed the convolution feature maps to f and g and did a matrix multiplication followed by a softmax function. Then the activated attention map times with the output of function h to get the self-attented feature maps.



2. Regular GAN's suffer from the vanishing gradient problem. An attempt to alleviate this issue is using Wasserstein loss. What is Wasserstein loss, and how is it different from the regular GAN loss function? Why does it not suffer as badly from vanishing gradients?

Answer: The Wasserstein loss compared with regular GAN loss is to use 1-Wasserstein distance to measure the distance of two distribution. In regular GAN, the distance metric is measured by Jensen-Shannon divergence where it will suffer from vanishing gradients if the two distributions are far away. But in Wasserstein distance, the metric between distributions are almost linear, meaning that it is easier to train the generator.

3. Find a GAN paper that interests you. Describe in detail the modifications it makes to regular GAN's, specifically in regards to its loss function and model architecture.

   Answer: The paper I chose is "Physics-Informed Generative Adversarial Networks for Stochastic Differential Equations". In this paper, the authors developed a new class of physical-informed GAN (PI-GAN) for solving stochastic differential equations. The model was developed on the top of Wassterstein-GAN with gradient penalty. Since there are more than one variable in a differential equation, the authors used 2 generators for approximating solution and parameters, 2 generators for approximating forcing terms as it shown in the figure below. The loss function for PI-GAN is the same as W-GAN where it uses the infimum of the expected value of wasserstein distance.

